Juniper Networks® vMX Release Notes

Release 18.2R2
21 July 2020

These release notes accompany this release of the Juniper Networks® virtual MX Series router (vMX). They describe new and changed features, limitations, and known problems in the software.

Contents

Introduction

The virtual MX Series router (vMX) is an MX Series router optimized to run on x86 servers. We recommend Ubuntu as the host OS.

vMX allows you to leverage Junos OS Release 18.2 to provide quick and flexible deployment. vMX provides the following benefits:

- Optimizes carrier-grade routing for the x86 environment
- Simplifies operations by consistency with MX Series routers
- Introduces new services without reconfiguration of current infrastructure

New and Changed Features

There are no new features in Junos OS Release 18.2R2 for the virtual MX Series router (vMX).

Known Behavior

This section contains the known behaviors and limitations in this release.

- When vMX is deployed, the management port is not configured, so you must use the serial console for configuration. Only a small number of configuration lines can be pasted in the vMX console. As a workaround, perform initial configuration to set the root password and to allow SSH access in the vMX console and perform further configuration using SSH.

Known Issues

This section lists the known issues in this release.

- The input **int** field of the MPLS version 4 data records reports the SNMP index value of the LSI interface instead of the ingress physical interface. PR1312047

- The Intel X710 NICs with 4 x 10G ports might observe a lower pulse per second (PPS) when all ports are running a 40G line rate. A 40G line rate can be achieved at a 512 byte packet size for the Intel X710 NIC. PR1281366

Resolved Issues

This section lists the resolved issues in this release.

- The TACACS access does not work after an upgrade. PR1220671
- The default adapter type changed from E1000 to VMXNET3. PR1365337
- Upgrade to Junos OS Release 18.1R1.9 fails and installing the **nfx-2-routing-data-plane-1.0-0.x86_64** package requires 76 MB on the file system. PR1366324
- The devices running VRRP on LAN are stuck in the master state. PR1371838
- When you configure the group VPN members using **authentication-method pre shared key ascii-text** the encryption and decryption do not work correctly due to the Packet Forwarding Engine discards. PR1381316
- On the vMX platform, a core file might be generated if the ifstate exception occurs. PR1398320
- On the vMX platform, the FPC might stop working. PR1364624

Upgrade Instructions

You cannot upgrade Junos OS for the vMX router from earlier releases using the **request system software add** command.

You must deploy a new vMX instance using the downloaded software package.

System Requirements

IN THIS SECTION

These topics provide system requirements for each supported environment.

Minimum Hardware and Software Requirements for KVM

Table 1 on page 4 lists the hardware requirements.

Table 1: Minimum Hardware Requirements for KVM

Description	Value
Sample system configuration	For lab simulation and low performance (less than 100 Mbps) use cases, any x86 processor (Intel or AMD) with VT-d capability. For all other use cases, Intel Ivy Bridge processors or later are required. Example of Ivy Bridge processor: Intel Xeon E5-2667 v2 @ 3.30 GHz 25 MB Cache For single root I/O virtualization (SR-IOV) NIC type, use Intel 82599-based PCI-Express cards (10 Gbps) and Ivy Bridge processors.

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Number of cores **NOTE:** Performance mode is the default mode and the minimum value is based on one port.	For lite mode with lab simulation use case applications: Minimum of 3 • 1 for VCP • 2 for VFP **NOTE:** If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.
	For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9 • 1 for VCP • 8 for VFP The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP. **NOTE:** To calculate the optimal number of vCPUs needed by VFP for performance mode: • Without CoS—(4 * *number-of-ports*) + 4 • With CoS—(5 * *number-of-ports*) + 4 **NOTE:** All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.

Memory
 	
NOTE: Performance mode is the default mode.
	For lite mode: Minimum of 3 GB
	
	• 1 GB for VCP
	• 2 GB for VFP
	
	For performance mode:
	
	• Minimum of 5 GB
	  1 GB for VCP
	  4 GB for VFP
	• Recommended of 16 GB
	  4 GB for VCP
	  12 GB for VFP
	
	Additional 2 GB recommended for host OS

Table 1: Minimum Hardware Requirements for KVM *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage
Other requirements	Intel VT-d capability Hyperthreading (recommended) AES-NI

Table 2 on page 6 lists the software requirements for Ubuntu.

Table 2: Software Requirements for Ubuntu

Description	Value
Operating system	Ubuntu 16.04.5 LTS (recommended host OS) Linux 4.4.0-generic
Virtualization	QEMU-KVM 2.0.0+dfsg-2ubuntu1.11
Required packages **NOTE:** Other additional packages might be required to satisfy all dependencies.	bridge-utils qemu-kvm libvirt-bin python python-netifaces vnc4server libyaml-dev python-yaml numactl libparted0-dev libpciaccess-dev libnuma-dev libyajl-dev libxml2-dev libglib2.0-dev libnl-dev python-pip python-dev libxml2-dev libxslt-dev **NOTE:** libvirt 1.3.1

Table 3 on page 6 lists the software requirements for Red Hat Enterprise Linux.

Table 3: Software Requirements for Red Hat Enterprise Linux

Description	Value
Operating system	• Red Hat Enterprise Linux 7.2 Linux 3.10.0-327.4.5 • Starting with Junos OS Release 17.3R1 Red Hat Enterprise Linux 7.3 Linux 3.10.0-514.6.2
Virtualization	QEMU-KVM 1.5.3

Table 3: Software Requirements for Red Hat Enterprise Linux *(continued)*

Description	Value
Required packages **NOTE:** SR-IOV requires these packages: kernel-devel gcc	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.17 or later

Table 4 on page 7 lists the software requirements for CentOS.

Table 4: Software Requirements for CentOS

Description	Value
Operating system	CentOS 7.2 Linux 3.10.0-327.22.2
Virtualization	QEMU-KVM 1.5.3
Required packages	python27-python-pip python27-python-devel numactl-libs libpciaccess-devel parted-devel yajl-devel libxml2-devel glib2-devel libnl-devel libxslt-devel libyaml-devel numactl-devel redhat-lsb kmod-ixgbe libvirt-daemon-kvm numactl telnet net-tools **NOTE:** libvirt 1.2.19 To avoid any conflicts, install libvirt 1.2.19 instead of updating from libvirt 1.2.17.

Minimum Hardware and Software Requirements for VMware

Table 5 on page 8 lists the hardware requirements.

Table 5: Minimum Hardware Requirements for VMware

<table>
<tr><th>Description</th><th>Value</th></tr>
<tr><td rowspan="2">Number of cores

NOTE: Performance mode is the default mode and the minimum value is based on one port.</td><td>For performance mode with low-bandwidth (virtio) or high-bandwidth (SR-IOV) applications: Minimum of 9

• 1 for VCP
• 8 for VFP

The exact number of vCPUs needed differs depending on the Junos OS features that are configured and other factors, such as average packet size. You can contact Juniper Networks Technical Assistance Center (JTAC) for validation of your configuration and make sure to test the full configuration under load before use in production. For typical configurations, we recommend the following formula to calculate the minimum vCPUs needed by the VFP.

NOTE: To calculate the optimal number of vCPUs needed by VFP for performance mode:

• Without CoS—(4 * number-of-ports) + 4
• With CoS—(5 * number-of-ports) + 4

NOTE: All VFP vCPUs must be in the same physical non-uniform memory access (NUMA) node for optimal performance. In addition to vCPUs for the VFP, we recommend 2 x vCPUs for VCP and 2 x vCPUs for Host OS on any server running the vMX.</td></tr>
<tr><td>For lite mode: Minimum of 3

• 1 for VCP
• 2 for VFP

NOTE: If you want to use lite mode when you are running with more than 3 vCPUs for the VFP, you must explicitly configure lite mode.</td></tr>
<tr><td>Memory

NOTE: Performance mode is the default mode.</td><td>For performance mode:

• Minimum of 5 GB
1 GB for VCP
4 GB for VFP
• Recommended of 16 GB
4 GB for VCP
12 GB for VFP

For lite mode: Minimum of 3 GB

• 1 GB for VCP
• 2 GB for VFP</td></tr>
</table>

Table 5: Minimum Hardware Requirements for VMware *(continued)*

Description	Value
Storage	Local or NAS Each vMX instance requires 30 GB of disk storage

Table 6 on page 9 lists the software requirements.

Table 6: Software Requirements for VMware

Description	Value
Hypervisor	ESXi 5.5 Update 2
Management Client	vSphere 5.5 or vCenter Server

Requesting Technical Support

Technical product support is available through the Juniper Networks Technical Assistance Center (JTAC). If you are a customer with an active J-Care or Partner Support Service support contract, or are covered under warranty, and need post-sales technical support, you can access our tools and resources online or open a case with JTAC.

- JTAC policies—For a complete understanding of our JTAC procedures and policies, review the *JTAC User Guide* located at https://www.juniper.net/us/en/local/pdf/resource-guides/7100059-en.pdf.
- Product warranties—For product warranty information, visit http://www.juniper.net/support/warranty/.
- JTAC hours of operation—The JTAC centers have resources available 24 hours a day, 7 days a week, 365 days a year.

Self-Help Online Tools and Resources

For quick and easy problem resolution, Juniper Networks has designed an online self-service portal called the Customer Support Center (CSC) that provides you with the following features:

- Find CSC offerings: https://www.juniper.net/customers/support/
- Search for known bugs: https://prsearch.juniper.net/
- Find product documentation: https://www.juniper.net/documentation/
- Find solutions and answer questions using our Knowledge Base: https://kb.juniper.net/
- Download the latest versions of software and review release notes: https://www.juniper.net/customers/csc/software/
- Search technical bulletins for relevant hardware and software notifications: https://kb.juniper.net/InfoCenter/
- Join and participate in the Juniper Networks Community Forum: https://www.juniper.net/company/communities/
- Create a service request online: https://myjuniper.juniper.net

To verify service entitlement by product serial number, use our Serial Number Entitlement (SNE) Tool: https://entitlementsearch.juniper.net/entitlementsearch/

Creating a Service Request with JTAC

You can create a service request with JTAC on the Web or by telephone.

- Visit https://myjuniper.juniper.net.
- Call 1-888-314-JTAC (1-888-314-5822 toll-free in the USA, Canada, and Mexico).

For international or direct-dial options in countries without toll-free numbers, see https://support.juniper.net/support/requesting-support/.

Revision History

21 July 2020—Revision 3, Junos OS Release 18.2R2—vMX

21 February 2019—Revision 2, Junos OS Release 18.2R2—vMX

15 December 2018—Revision 1, Junos OS Release 18.2R2—vMX